

playlist guitar lessons

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Project Spec

Web app to convert Spotify playlists into skill trees.

- will crawl the internet for chord sheets and or tabs
- generate a 'skill tree' with different branches for different techniques (e.g. harder chords and different strumming patterns)



User Stories



As a beginner guitar player, I want to filter low difficulty songs tabs/chords, so that I can find songs that I can play at my current level, and continue with songs that I can play as I improve.

User Stories



As a guitar teacher, I want to give my students a system where they can upload their own songs/playlists and get their tabs, so that my students can engage more with their learning.

User Stories



As an intermediate guitar player, I want a way to filter song tabs by skills used, so that I can improve whatever skill I want to target.

User Stories



As a guitar player who can't afford a teacher, I want a system with sections for difficulty of song tabs/chords, so that it can walk me through a useful learning journey.

The MVP Plan

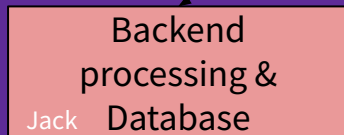
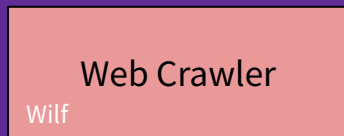
- Takes a list of songs and sends them to web scraper
- Scrapes web to return list of chords/tabs
- An algorithm to order tabs by difficulty
- A frontend that can present said information
- Some level of account integration
- Local hosting

The resulting MVP

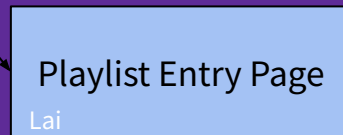
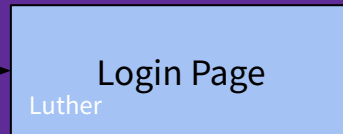
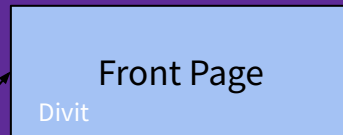
- Takes a list of songs and sends them to web scraper
- Scrapes web to return list of chords/tabs
- An algorithm to order tabs by difficulty
- A frontend that can present said information - We had a few issues hooking up the frontend and backend so data isn't sent back properly just yet
- Some level of account integration - Database out of scope, wanted to focus on functionality
- Local hosting

Setting out

Backend:



Frontend:



STACK

FRONTEND

 TypeScript

 React

 Vite

WEB SERVER

 caddy®

BACKEND

 TypeScript

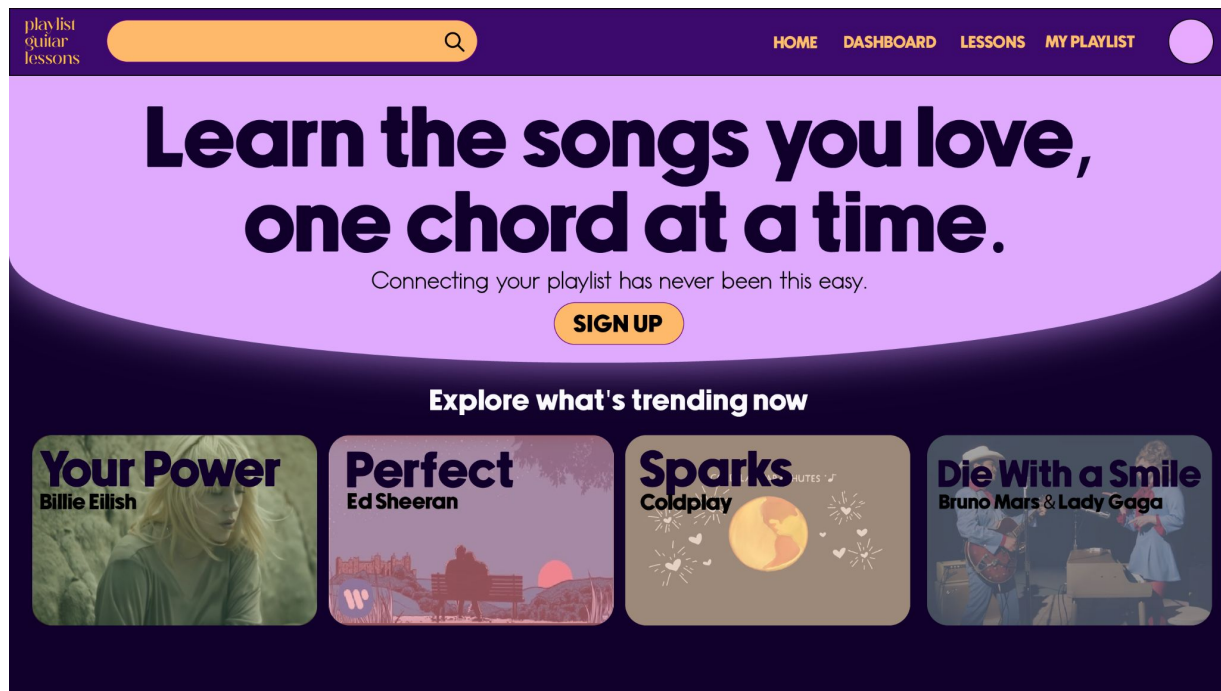
 nest

DATABASE

 GraphQL

INITIAL WEBSITE DESIGN

HOMEPAGE



LOGO VARIATIONS

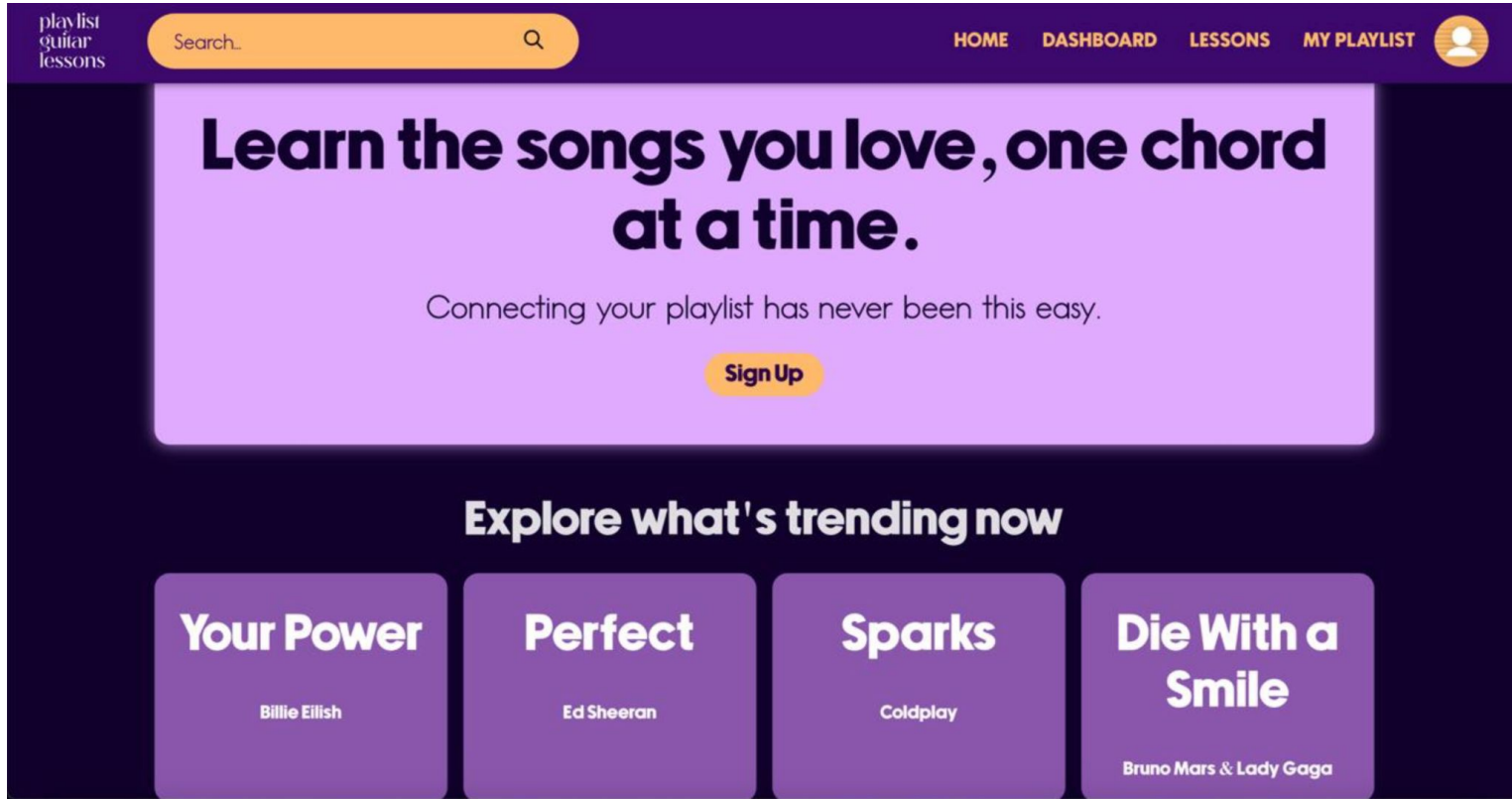


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First Implementation Attempt...



Controller

Description

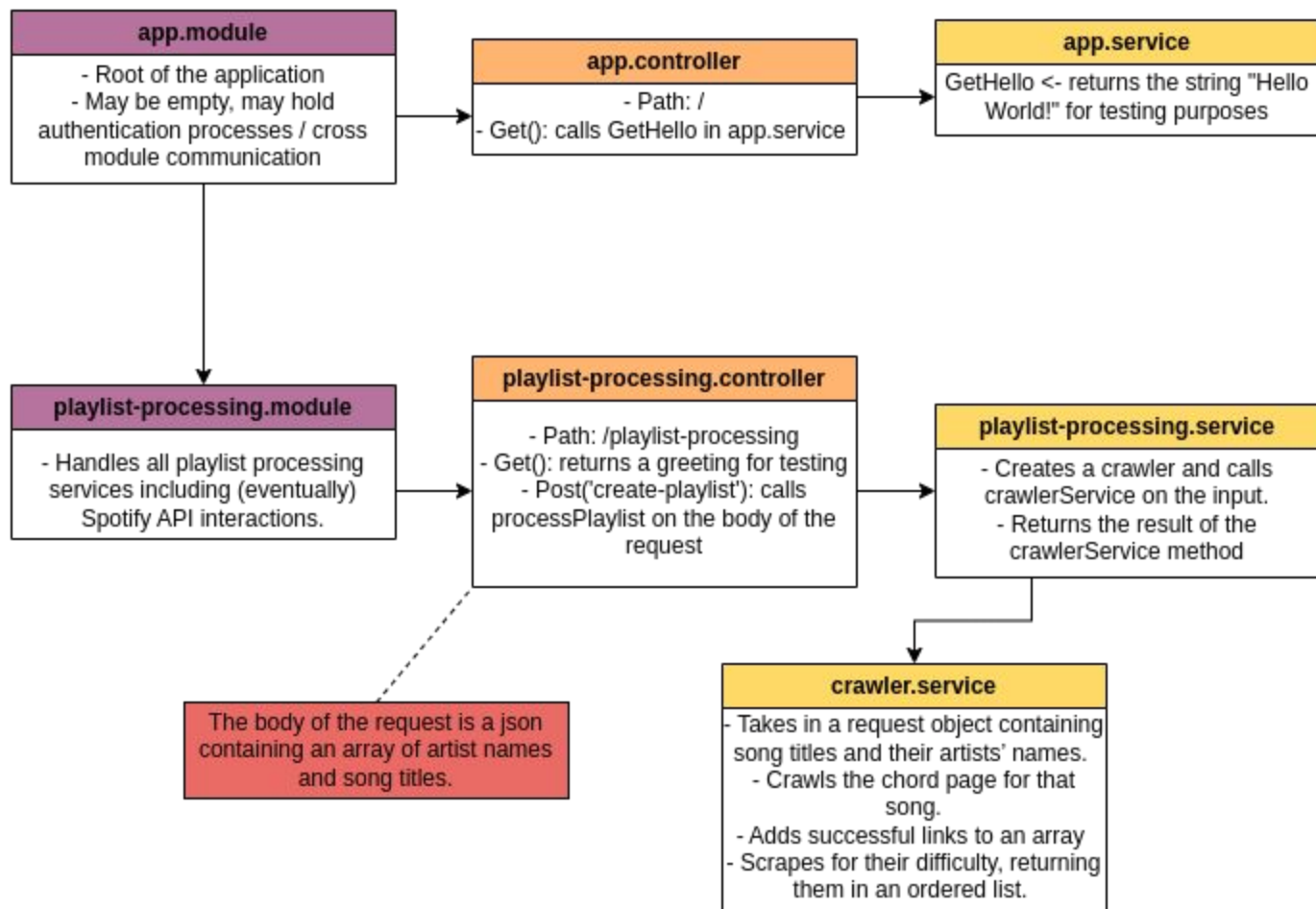
Module

Description

Service

Description

Comment



Crawler

```
1 {  
2   "playlist" : [  
3     {  
4       "title": "letter to an old poet" ,  
5       "artist": "boygenius"  
6     },  
7     {  
8       "title": "hallelujah" ,  
9       "artist": "leonard cohen"  
10    },  
11    {  
12      "title": "just like heaven" ,  
13      "artist": "the cure"  
14    },  
15    {  
16      "title": "random test song" ,  
17      "artist": "literally nobody"  
18    },  
19    {  
20      "title": "your power" ,  
21      "artist": "billie eilish"  
22    }  
23  ]  
24 }  
25
```



```
here are the successful links sorted by difficulty :  
[  
  [ 'https://www.mychordbook.com/chords/leonard-cohen/hallelujah', 5 ],  
  [ 'https://www.mychordbook.com/chords/boygenius/letter-to-an-old-poet',  
    6  
  ],  
  [ 'https://www.mychordbook.com/chords/the-cure/just-like-heaven', 6 ]  
]
```

First we have to get the links

Search results for: just like heaven

2 results



Just Like Heaven
Katie Melua



Just Like Heaven
The Cure



BROWSE ALL ARTISTS

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z 123

Success!

We can just do this for every link

mychordbook.com

The Cure - Just Like Heaven Chords

Search song or artist

CHORDS PRICING

The Cure - Just Like Heaven Chords

DO NOT SHOW ADS

Intro:
( E Bm D) 8x

Show me, show me, show me how you do that trick
Bm D

The one that makes me scream she said
A E

The one that makes me laugh she said
Bm D

And threw her arms around my neck

A E

Show me how you do it
Bm D

And I promise you I promise that

I'll run away with you
A E

I'll run away with you
Bm D

A | la major

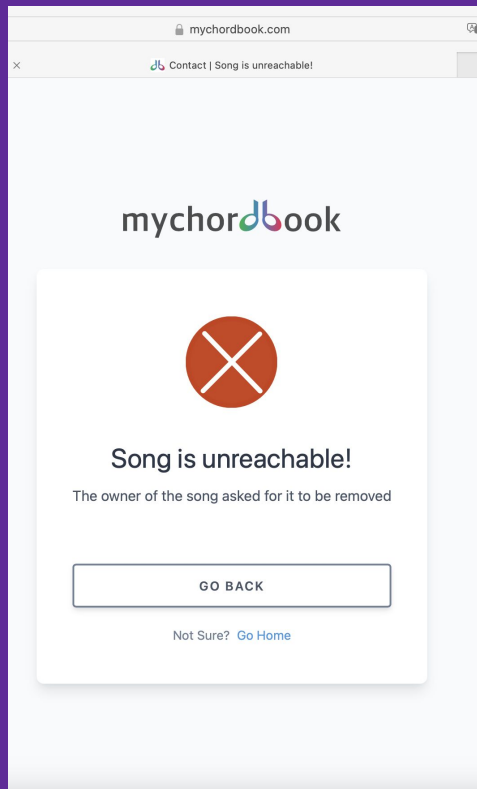
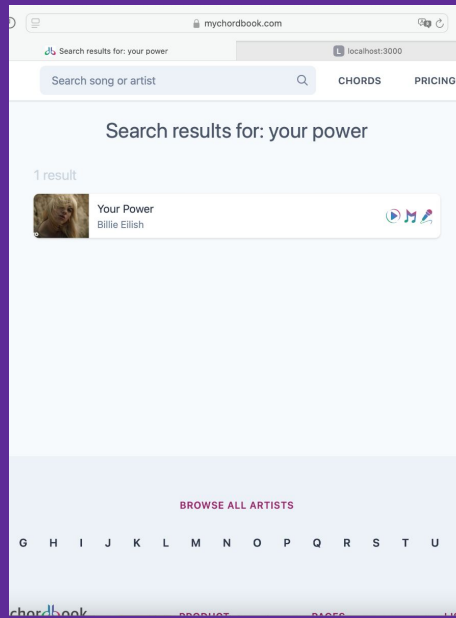
NO CAPO

3
2
1

3 5 7

swimming on that dizzy edge

But not always...



We need to check each link individually via an http request and check we get html from it

With error checking for
broken links and
nonexistent songs...

the successful links are as follows :

```
[  
  'letter to an old poet',  
  'https://www.mychordbook.com/chords/boygenius/letter-to-an-old-poet'  
]  
[  
  'hallelujah',  
  'https://www.mychordbook.com/chords/leonard-cohen/hallelujah'  
]  
[  
  'just like heaven',  
  'https://www.mychordbook.com/chords/the-cure/just-like-heaven'  
]
```

the unsuccessful links are as follows :

```
[ 'random test song', "sorry, can't find that song" ]  
[ 'your power', "can't reach due to server error" ]
```

Now we rank 'em

The Cure - Just Like Heaven Chords

DO NOT SHOW ADS

Intro:
(A E Bm D) 8x

Show me, show me, show me how you do that trick
Bm D

The one that makes me scream she said
A E

The one that makes me laugh she said
Bm D

And threw her arms around my neck

A E
Show me how you do it
Bm D

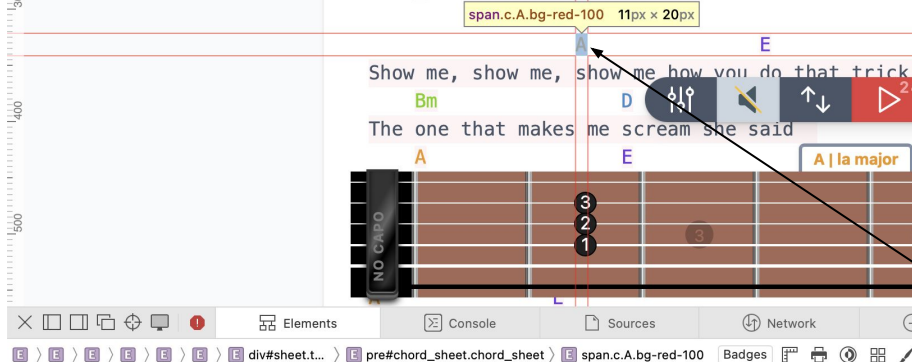
And I promise you I promise
A E

I'll run away with you
A | la major

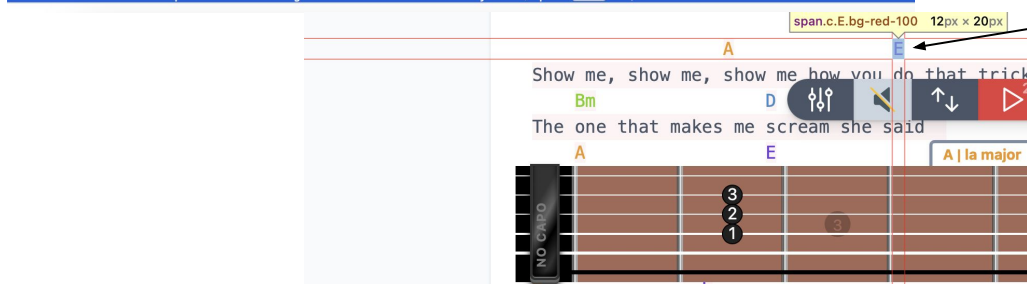
The Cure - Just Like H...
0:00 / 3:27
vevo
YouTube

PEOPLE ALSO VIEWED

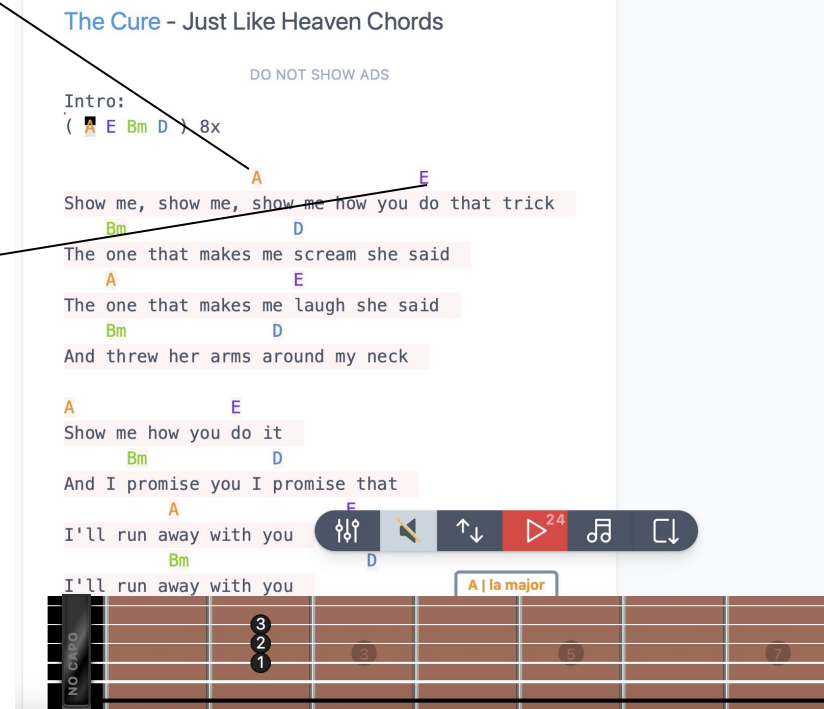
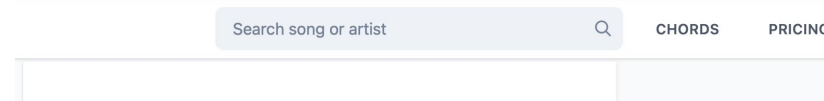
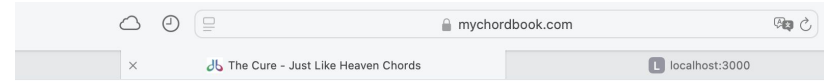
- So Far Away
Averged Sevenfold
- A Forest
The Cure
- Boys Dont Cry
The Cure
- Let
Pinegrove
- Alaska
Pinegrove
- These Are The Days
Cory Asbury
- My Inheritance



```
<pre class="chord_sheet" id="chord_sheet">
  <span class="l bold" data-id="0" tabindex="-1" style="outline: currentcolor;">...</span> Event
  " ( "
  <span class="c A" data-id="0" style=A</span> Event
  <span class="chord_index" style="display: none;">0</span>
  <span class="c E" data-id="1" style="background: -webkit-linear-gradient(top, black
  1.328022115059668%, #e4e4e4 0%);">E</span> Event
  <span class="chord_index" style="display: none;">1</span>
  <span class="c Bm" data-id="2" style=Bm</span> Event
  <span class="chord_index" style="display: none;">2</span>
  <span class="c D" data-id="3" style=D</span> Event
  <span class="chord_index" style="display: none;">3</span>
  " ) 8x
  <span class="l1" data-id="1">...</span> Event
  <span class="c A bg-red-100" data-id="4" style=A</span> Event = $0
```



```
<pre>
  <span class="c B" data-id="2" style=Bm</span> Event
  <span class="chord_index" style="display: none;">2</span>
  <span class="c D" data-id="3" style=D</span> Event
  <span class="chord_index" style="display: none;">3</span>
  " ) 8x "
  <span class="l1" data-id="1">...</span> Event
  <span class="c A bg-red-100" data-id="4" style=A</span> Event
  <span class="chord_index" style="display: none;">4</span>
  <span class="c E bg-red-100" data-id="5" style=E</span> Event = $0
```



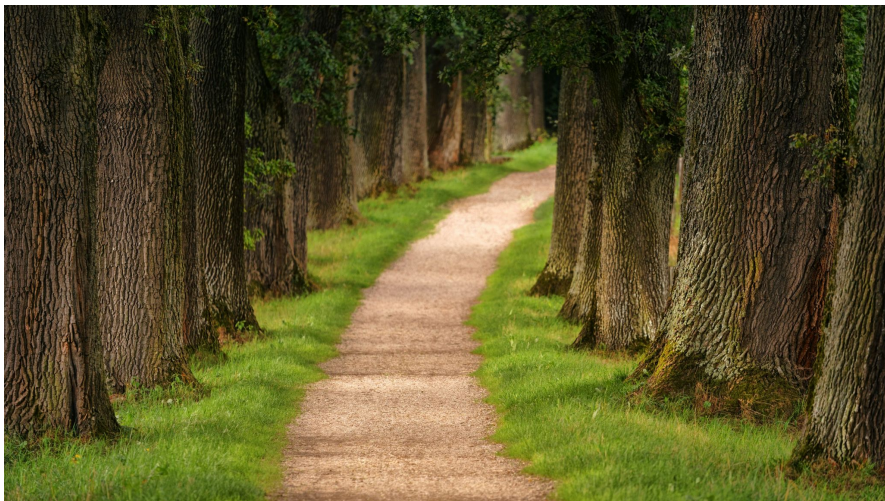
Chord array of the successful links:

```
[ 'F', 'C', 'G', 'Am', 'Em', 'Dm' ]  
[ 'C', 'G', 'Am', 'F', 'E7' ]  
[ 'A', 'E', 'Bm', 'D', 'F#m', 'G' ]  
[
```

here are the successful links sorted by difficulty :

```
[  
  [ 'https://www.mychordbook.com/chords/leonard-cohen/hallelujah', 5 ],  
  [  
    'https://www.mychordbook.com/chords/boygenius/letter-to-an-old-poet',  
    6  
  ],  
  [ 'https://www.mychordbook.com/chords/the-cure/just-like-heaven', 6 ]  
]
```

Where to next...



- a) Getting proper front/backend integration
 - b) Spotify Integration
 - c) Database setup
 - d) Account integration
 - e) Robustness for the crawler (don't just hardcode for one site)
 - f) More comprehensive ranking of songs (more than just no. chords)
 - g) Closer working with one another
 - h) Continuous Integration
-

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Any Questions ?